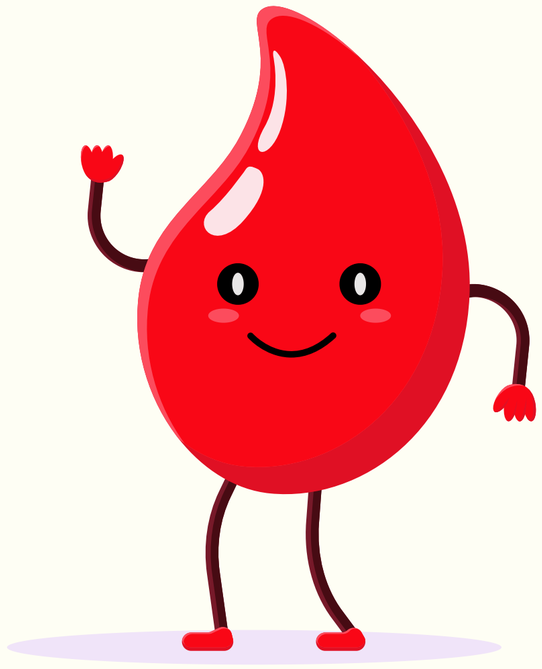


# BLOODLINK

## SMART EMERGENCY BLOOD DONATION PLATFORM

Connecting Lives Through Real-Time Blood Donation.

**UX Research Case Study by Anees Kumar Ghosh**

# Project Overview

- BloodLink is a real-time blood donation platform that connects verified blood donors with people in urgent need of blood.
- Using location-based matching, the app quickly identifies nearby eligible donors, reducing response time during emergencies.
- It provides a simple, secure, and reliable way to request and donate blood, helping save lives through faster coordination.



# Problem Statement

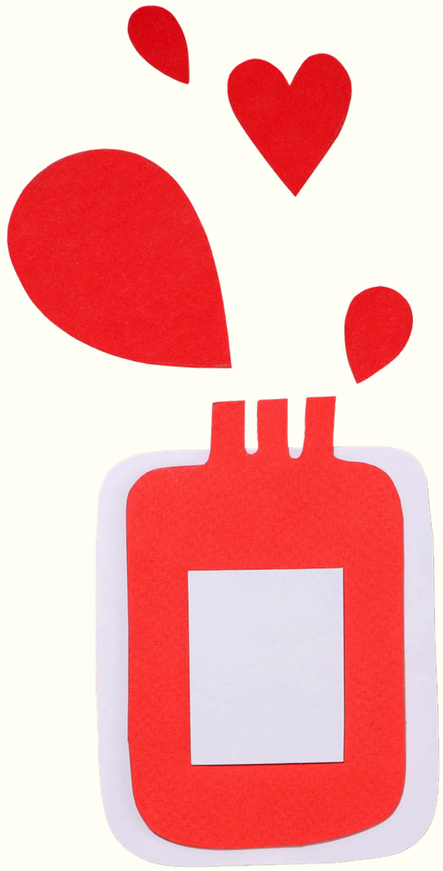
- Finding blood during emergencies is often time-consuming and unorganized.
- Patients depend on manual communication through hospitals, NGOs, and social media, while outdated donor records and limited blood availability lead to treatment delays.



# Solution

- BloodLink is a location-based emergency blood donation platform.
- Connects patients with nearby verified donors in real time.
- Users can instantly create emergency blood requests.
- The system identifies eligible donors based on: Blood group, Location, Donation eligibility.
- Live tracking,Instant notifications,Secure communication;help reduce response time.
- Simplify the donation process.

# Research



- Understand the current blood request process.
- Identify pain points in existing systems.
- Analyze how hospitals manage blood shortages.
- Understand donor motivation and behavior.
- Discover opportunities for technology intervention.
- Validate the need for a real-time donor matching platform.

# User Interviews

## Donors

- Rarely receive verified requests.
- Random calls reduce trust.
- Unsure about eligibility.

## NGO Volunteers

- Donor lists become outdated quickly.
- Calls and messaging apps are time-consuming.
- Tracking donor availability is difficult.

## Patients

- High stress during emergencies.
- Difficulty locating compatible donors.
- Dependence on hospitals & social media.

## Hospital Staff

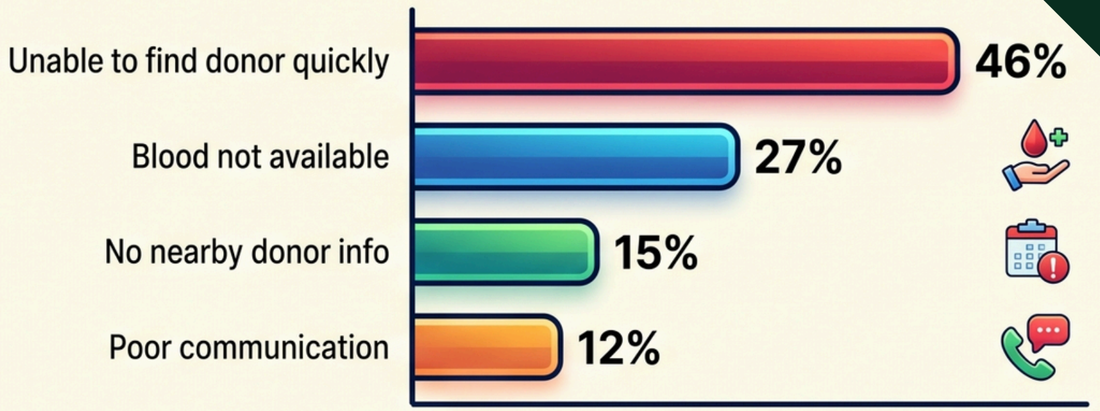
- Blood stock is unpredictable.
- Manual donor contacting is time-consuming.
- Existing processes are fragmented.

# Survey Results

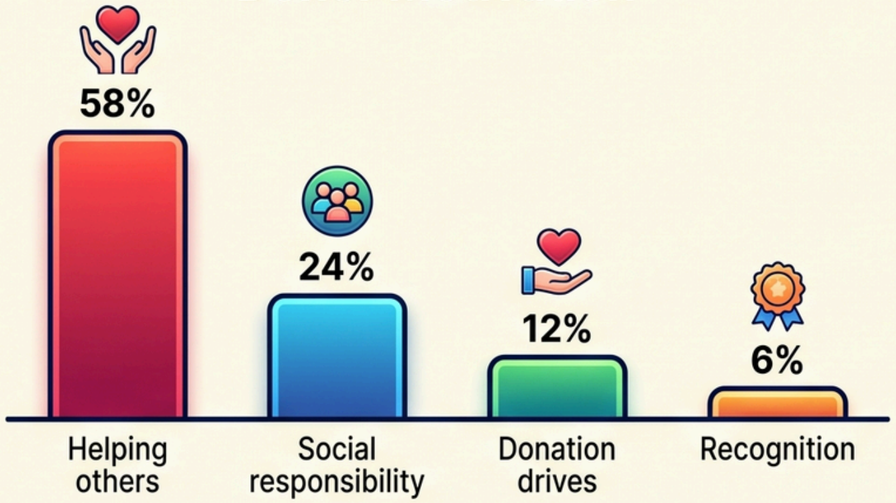
## How do people search for blood?



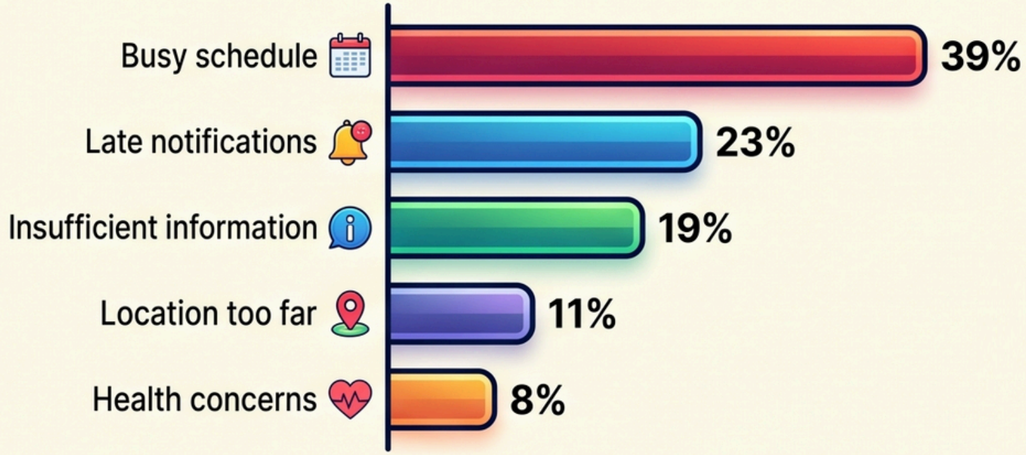

## Biggest Challenges




## Donor Motivation




## Reasons Donors Ignore Requests



# Current Workflow

1. Patient needs blood
2. Family contacts hospital
3. Hospital checks inventory
4. If unavailable, blood bank is contacted
5. NGOs are informed
6. Messages shared via WhatsApp / Facebook
7. Volunteers call donors
8. Suitable donor is found
9. Donor travels to hospital
10. Blood is donated

**Average Response Time:** 2–12 Hours

# Existing Problems

## Patients

- Slow response time
- No centralized platform
- Limited visibility of nearby donors

## Hospitals

- Manual coordination
- Limited inventory
- No real-time donor info

## Donors

- Fake or incomplete requests
- No availability management
- Random calls

## NGOs

- Outdated donor databases
- Manual communication
- No analytics

# Competitor Analysis

## e-RaktKosh

✓ **Inventory management**

✗ Limited real-time donor matching

## Red Cross Blood Services

✓ **Donation campaigns**

✗ Limited emergency peer-to-peer matching

## Generic Blood Donor Apps

✓ **Donor registration**

✗ Poor verification

# User Personas

## EMERGENCY PATIENT



**Age: 30** Office Employee

**GOAL**

Find compatible blood within one hour.

**PAIN POINTS**

- No trusted donor contacts
- Hospital has limited stock
- Manual searching is stressful

## BLOOD DONOR



**Age: 24** Student

**GOAL**

Donate blood safely when needed.

**PAIN POINTS**

- Receives incomplete requests
- Unsure about eligibility
- Wants verified communication

## HOSPITAL COORDINATOR



**Age: 38** Emergency Coordinator

**GOAL**

Arrange blood quickly for critical patients.

**PAIN POINTS**

- Limited blood inventory
- Manual donor coordination
- No centralized real-time system

# Conclusion

- The research revealed that the primary challenge is not the shortage of willing donors, but the lack of an efficient, trusted, and real-time coordination system.
- Existing processes rely heavily on manual communication, causing delays during emergencies.
- BloodLink addresses this gap by introducing:

1. Verified donor profiles
2. Intelligent location-based matching
3. Real-time notifications
4. Hospital collaboration
5. Transparent request tracking

- Creating a faster and more reliable emergency blood donation ecosystem.

# Thank You

